

Aerosol-Cloud Interactions in ERF

A practical experiment and development plan using bulk microphysics and SHOC

Technical note based on the ERF source tree reviewed 31 August 2026

ABSTRACT

This note describes how to investigate aerosol-cloud interactions with the microphysics schemes currently available in ERF. Morrison and WDM6 provide the most direct bulk-microphysics pathways: Morrison accepts a prescribed cloud droplet number, while WDM6 carries cloud condensation nuclei (CCN), cloud droplet number, and rain number. Kessler, SAM, and WSM6 are useful for controlled process-sensitivity studies but do not currently represent aerosol activation explicitly. SHOC adds the turbulence and cloud-macrophysics dimension, although its native state-update coupling presently lacks the number closure required by Morrison and WDM6. A staged program is recommended to separate aerosol effects, uncertain warm-rain process rates, turbulent mixing, and radiation coupling.

1. Main conclusions

The scientifically cleanest near-term program is to use Morrison-NoIce and WDM6 for warm-cloud aerosol perturbations, with identical dynamics, forcing, resolution, and surface conditions. Morrison supplies a controlled prescribed-droplet-number experiment. WDM6 is the more literal bulk aerosol experiment because it predicts CCN, cloud-droplet number, and rain number.

Kessler, SAM, and WSM6 can still answer important questions about cloud and precipitation susceptibility. Their autoconversion thresholds, collection efficiencies, and accretion rates should be interpreted as microphysical process uncertainty or aerosol proxies, not as explicit aerosol activation.

SHOC is not an aerosol model. It changes the thermodynamic variance, subgrid-cloud diagnosis, mixing, and entrainment through which aerosol perturbations operate. The ideal target is therefore a factorial experiment combining aerosol level with SHOC closure settings. In the current native SHOC implementation, however, number-aware microphysics requires an additional closure before that full experiment is possible.

2. Physical framing

The chain to resolve is:

**Aerosol/CCN -> activation -> droplet number and size
-> autoconversion/accretion -> precipitation
-> cloud water, cloud fraction, lifetime, and radiation**

SHOC acts mainly on activation conditions, cloud fraction, condensate, and entrainment through its representation of subgrid variability and turbulent mixing. Bulk-microphysics tuning acts mainly on the conversion from condensate to precipitation. Separating these links is essential: changing an accretion coefficient can reproduce a precipitation response, but it is not equivalent to changing aerosol concentration.

3. Capabilities of the available schemes

Scheme	Current aerosol-related representation	Recommended role
Kessler	Warm-rain, single moment; predicts water vapor, cloud water, and rain-water only.	Autoconversion and accretion process sensitivity; aerosol proxy only.
SAM	Single-moment mixed-phase scheme with fixed thresholds, intercepts, and collection efficiencies.	Process uncertainty for liquid, ice, snow, and graupel conversion.
WSM6	Single-moment mixed phase; cloud-droplet number is a fixed internal assumption, $x_{ncr} = 3e8 \text{ m}^{-3}$ (300 cm^{-3}).	Expose x_{ncr} as a runtime input for prescribed-droplet-number proxy experiments.
Morrison	Two-moment framework, but ERF currently uses constant cloud-droplet number selected by <code>erf.morrison_ndcnst</code> .	Best no-code prescribed-Nc experiment; use Morrison-NoIce first for warm clouds.
WDM6	Predicts CCN aerosol (nn), cloud-droplet number (nc), and rain number (nr); ice remains single moment.	Best existing bulk scheme for explicit aerosol-cloud interaction experiments.
SuperDroplets	Represents aerosol size/composition, activation, condensation, stochastic collision-coalescence, and sedimentation with particles.	Higher-fidelity benchmark for a limited subset of bulk experiments.

3.1. Kessler

Kessler has no aerosol or number-concentration variable. Its warm-rain conversion currently uses a cloud-water autoconversion threshold of $1e-3 \text{ kg kg}^{-1}$, an autoconversion coefficient of $1e-3 \text{ s}^{-1}$, and an accretion term proportional to $2.2 q_c q_r^{0.875}$. These values are compiled constants.

Useful additions would be scheme-local runtime inputs for the autoconversion threshold, autoconversion multiplier, and accretion multiplier. Defaults should exactly preserve the present solution. Such experiments quantify structural sensitivity of precipitation formation; they should not be labeled direct aerosol experiments.

3.2. SAM

SAM is a single-moment mixed-phase scheme. It shares fixed liquid and ice autoconversion thresholds and uses fixed rain, snow, and graupel distribution intercepts and collection efficiencies. Recommended tuners are liquid and ice autoconversion thresholds, conversion-rate multipliers, and collection- efficiency multipliers. These are particularly useful for assessing whether an apparent aerosol signal is robust to uncertainty in ice and precipitation processes.

3.3. WSM6

WSM6 assumes $x_{ncr} = 3e8 \text{ m}^{-3}$, equivalent to 300 cm^{-3} , inside the scheme. A runtime `wsm6.xncr` option would enable clean, intermediate, and polluted proxy experiments. Coefficients derived from x_{ncr} must be recomputed consistently in both the native C++ and Fortran paths. Reasonable first levels are 50, 300, and 1000 cm^{-3} , followed by a denser sweep if the response is nonlinear.

Higher prescribed droplet number generally decreases droplet size and delays warm-rain formation in this class of parameterization. It remains a prescribed single-moment assumption: aerosol transport, activation, depletion, and regeneration are absent.

3.4. Morrison

ERF exposes the constant droplet concentration through `erf.morrison_ndcnst`, with a default of 250 cm^{-3} . This is immediately suitable for a prescribed-Nc ensemble. Begin with Morrison-NoIce to isolate the warm- cloud pathway, using 50, 250, and 1000 cm^{-3} as clean, intermediate, and polluted conditions. Repeat selected cases with full Morrison to evaluate mixed-phase dependence.

This experiment alters droplet size and the number-dependent warm-rain pathway, but it still does not calculate activation from a prognostic aerosol population. The interpretation is therefore a cloud-droplet-number susceptibility, not a complete aerosol life-cycle simulation.

3.5. WDM6

WDM6 is the strongest current bulk option for aerosol-cloud interaction. It uses a runtime background CCN concentration, `wdm6.ccn0`, and carries `nn`, `nc`, and `nr` along with the hydrometeor mass fields. A useful first sweep is $1e8$, $5e8$, and $1e9 \text{ m}^{-3}$, equivalent to 100, 500, and 1000 cm^{-3} .

There is an important implementation caveat. At every macro timestep, `nn` is clamped into the interval $1e8$ to $2e10 \text{ m}^{-3}$. Values below $1e8 \text{ m}^{-3}$ are therefore ineffective, and activation depletion below the floor is replenished. The current `ccn0` should be described as a maintained background reservoir rather than a fully conserved aerosol population. The land/water selection also affects the autoconversion regime, so `erf.is_land` and `wdm6.hail_opt` must remain fixed across the aerosol ensemble.

3.6. SuperDroplets as a benchmark

The SuperDroplets option can specify dry aerosol number, size distribution, and composition and then represent activation and collision-coalescence explicitly. It is more expensive and statistically noisy, but a small matched set is valuable for determining where Morrison and WDM6 bulk assumptions control the result. Match aerosol number and dry-size distribution as closely as possible, and use multiple particle realizations.

4. Using SHOC to study turbulence-aerosol coupling

The most ACI-relevant shared SHOC controls are listed below. Change them one at a time by approximately plus or minus 25 percent around the default before attempting a multi-parameter ensemble.

Input	Default	ACI interpretation
<code>erf.shoc.qw2tune</code>	1.0	Controls total-water variance used by the sub-grid closure; influences cloud occurrence and condensate.
<code>erf.shoc.qwthl2tune</code>	1.0	Controls water-temperature covariance and therefore the joint thermodynamic PDF.
<code>erf.shoc.thl2tune</code>	1.0	Controls liquid-water-potential-temperature variance and saturation variability.
<code>erf.shoc.w2tune</code>	1.0	Controls vertical-velocity variance, relevant to updraft variability and activation conditions.
<code>erf.shoc.length_fac</code>	0.5	Changes the turbulent length scale and consequently mixing and entrainment.
<code>erf.shoc.coeff_kh</code>	0.1	Changes scalar and heat diffusivity; useful as a secondary mixing sensitivity.

Enable `erf.shoc.extra_shoc_diags` and save `shoc_cldfrac`, `shoc_ql`, `shoc_ql2`, `shoc_cond`, `wqls_sec`, `w_sec`, `thl_sec`, `qw_sec`, and `qwthl_sec` where available. These diagnostics distinguish a turbulence-mediated condensate response from a change produced directly by precipitation conversion.

4.1. Current coupling restriction

Native SHOC in moist simulations must use `state_update` because moist `host_diffusion` does not own the required cloud macrophysics. The native state-update driver currently rejects layouts containing cloud- or ice-number concentrations: changing cloud mass without a corresponding number tendency would leave the size distribution inconsistent. Thus native SHOC currently supports Kessler, SAM, and WSM6 more naturally than Morrison or WDM6.

For Morrison with prescribed `Nc`, a first closure could consistently reimpose the prescribed cloudy-cell `Nc` and clear it when cloud water vanishes. For WDM6, a defensible closure must transfer number between `nn` and `nc` during activation and return or remove number consistently during evaporation. This should be implemented and tested before interpreting a WDM6-by-SHOC factorial ensemble.

5. Recommended experiment program

5.1. Cases and controls

Use RICO as the primary precipitating warm-cumulus case. It exercises the aerosol-to-warm-rain pathway while remaining interpretable. Use DYCOMS as a secondary stratocumulus case for cloud-water and lifetime responses, and BOMEX as a weakly precipitating control. ERF already contains canonical cases for these configurations.

For every comparison, hold the grid, timestep, forcing, sounding, radiation, surface fluxes, land/water mask, microphysics substepping, and output cadence fixed. Use at least three initial-perturbation realizations when feasible; a single deterministic LES difference can be dominated by chaotic sampling.

5.2. Stage 1: no-code aerosol and droplet-number experiments

Configuration	Perturbation levels	Primary interpretation
Morrison-NoIce	morrison_ndcnst = 50, 250, 1000 cm⁻³	Prescribed cloud-droplet-number susceptibility.
WDM6	ccn0 = 1e8, 5e8, 1e9 m⁻³	Bulk CCN activation and warm-rain response, subject to the nn floor.
SuperDroplets subset	Matched clean and polluted aerosol distributions	Benchmark for activation and collision-coalescence assumptions.

5.3. Stage 2: process attribution

Add runtime multipliers for autoconversion and accretion, initially preserving all present defaults. Perturb one rate at a time, then use a compact factorial design around the aerosol levels. The goal is to determine whether the aerosol response exceeds uncertainty in the bulk conversion formulae and whether the sign of the response is robust.

Add budget diagnostics for activation (nn to nc), autoconversion (qc to qr), accretion of cloud water by rain, rain evaporation, sedimentation, and any ice-phase conversions used in the selected case. Record mass tendencies in kg kg⁻¹ s⁻¹ and number tendencies in kg⁻¹ s⁻¹, with column-integrated counterparts when possible.

5.4. Stage 3: SHOC interaction experiment

Before number closure is available, use native SHOC with WSM6 after xncr is made configurable, treating xncr as an aerosol proxy. Once number closure is implemented, repeat the Morrison and WDM6 clean/polluted end-points with the baseline and selected perturbed SHOC settings.

For any outcome Y, quantify the turbulence modulation with the difference-in-differences interaction:

$$I = (Y_{polluted,SHOC} - Y_{clean,SHOC}) - (Y_{polluted,noSHOC} - Y_{clean,noSHOC}).$$

A nonzero I shows that the aerosol response depends on the turbulence and subgrid-cloud treatment rather than simply adding to it.

5.5. Stage 4: radiation coupling

The present RRTMGP interface diagnoses cloud fraction as either zero or one in each grid cell and assigns a fixed liquid effective radius of 10 micrometers (and ice radius of 25 micrometers). It does not use predicted Nc or native SHOC cloud fraction. Therefore existing experiments can capture cloud amount, water-path, lifetime, and precipitation responses, but not a complete Twomey effect.

Before claiming aerosol-cloud-radiation interaction, pass a microphysics- derived liquid effective radius and the SHOC subgrid cloud fraction/in-cloud water to radiation. The mass-number estimate

$$r_v = \left[\frac{3q_c}{4\pi\rho_w N_c} \right]^{1/3}$$

provides a volume-mean radius when qc and Nc use compatible per-dry-air-mass units. Effective radius additionally requires a spectral-shape factor consistent with the microphysics size distribution.

6. Diagnostics and evaluation metrics

Save three-dimensional qv, qc, qi, qrain, qsnow, and qgraup as applicable; number concentrations nc, nr, nn, ni, ns, and ng where available; accumulated surface precipitation; and SHOC thermodynamic and turbulence diagnostics. Construct the following analysis products:

- Liquid and ice water paths, cloud fraction, cloud-base and cloud-top height, and cloud lifetime.
- Surface precipitation rate, accumulated precipitation, onset time, and precipitating area fraction.
- Cloud-droplet and rain-number distributions, droplet-size estimates, and joint qc-Nc histograms.

- Domain-integrated microphysical process budgets and their vertical profiles.
- Aerosol susceptibility, such as $d \ln(\text{LWP})/d \ln(N)$ and $d \ln(\text{precipitation})/d \ln(N)$, with ensemble uncertainty.
- Energy and water closure, including precipitation and sedimentation fluxes.

Compare time windows only after a common spin-up and report ensemble means, spread, and temporal auto-correlation-aware uncertainty. Avoid interpreting a single domain-mean difference without inspecting changes in the cloud population distribution.

7. Implementation priorities

1. Expose scheme-local Kessler and SAM conversion/collection parameters and WSM6 xncr as runtime inputs while preserving current defaults exactly.
2. Add named microphysical process-rate diagnostics and mass/number budget tests.
3. Implement and validate number closure for native SHOC with Morrison and WDM6.
4. Revisit WDM6's per-timestep CCN floor so conserved and maintained-reservoir experiments can be selected explicitly.
5. Couple microphysics-derived effective radius and SHOC cloud fraction to RRTMGP before evaluating the first aerosol indirect radiative effect.

8. Example ERF inputs

8.1. Morrison prescribed-droplet-number ensemble

```
erf.moisture_model = Morrison_NoIce
erf.morrison_ndcnst = 50.0 # repeat: 250.0, 1000.0 cm^-3
erf.plot_vars_1 = qv qc grain nc nr rain_accum
```

8.2. WDM6 background-CCN ensemble

```
erf.moisture_model = WDM6
wdm6.ccn0 = 1.0e8 # repeat: 5.0e8, 1.0e9 m^-3
wdm6.hail_opt = 0
erf.is_land = 0 # maritime idealized case; keep fixed
erf.plot_vars_1 = qv qc grain nc nn nr rain_accum
```

8.3. Native SHOC with a single-moment proxy scheme

```
zlo.type = surface_layer
erf.pbl_type = NATIVE_SHOC
erf.moisture_model = WSM6
erf.shoc.transport_mode = state_update
erf.shoc.extra_shoc_diags = true
```

This SHOC configuration is not an aerosol ensemble until WSM6 xncr is exposed and varied. Request the SHOC diagnostic plot variables explicitly in addition to enabling their calculation.

9. Code and documentation references

Paths are relative to the ERF repository root.

Constants and single-moment schemes: Source/ERF_Constants.H; Source/Microphysics/Kessler/ERF_KesslerUtils.H; and Source/Microphysics/WSM6/ERF_WSM6.H.

Morrison and WDM6: Source/Microphysics/Morrison/ERF_AdvanceMorrison.cpp; Source/Microphysics/WDM6/ERF_InitWDM6.cpp; and Source/Microphysics/WDM6/ERF_AdvanceWDM6.cpp.

SHOC and radiation: Source/PBL/Shoc/ERF_ShocDriver.H and Source/PhysicsInterfaces/Radiation/ERF_Radiation.cpp.

Documentation: Docs/sphinx_doc/Inputs.rst; Docs/sphinx_doc/theory/Microphysics.rst; and Docs/sphinx_doc/plotfiles/Plotfile3DReference.rst.